MACHINE LEARNING

House price prediction

Feature engineering, PCA, and model stacking on 1,460 homes

The problem

1,460
homes in the training data

81
columns, 80 of them describing the home and its lot

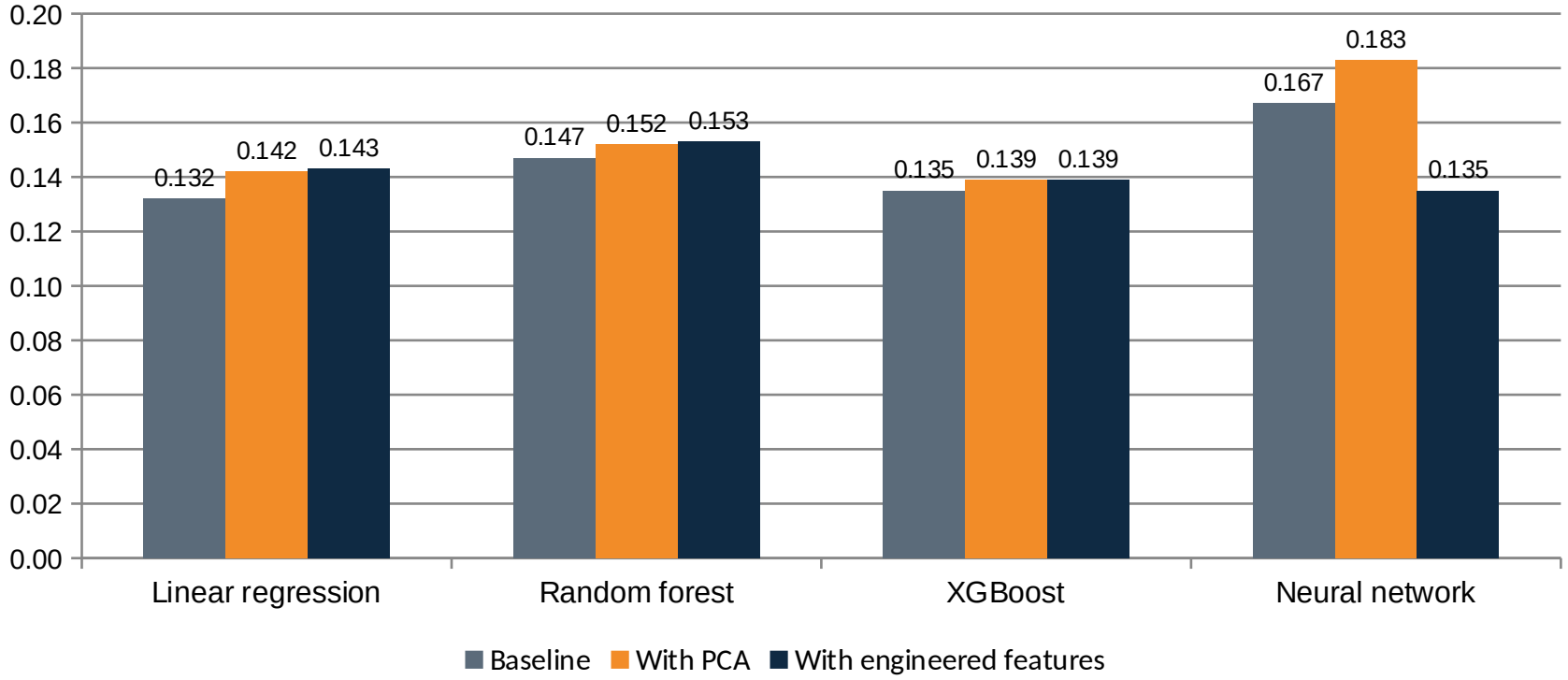
log
price is predicted on a log scale, scored by RMSE (lower is better)

- The goal is to predict each home’s sale price from its features
- Housing data is messy: missing values, skewed prices, and many categories

The workflow

1. Explore the data with seven questions about price, zoning, age, and living area
2. Build one preprocessing pipeline: fill gaps, scale numbers, one-hot encode categories
3. Tune four models with grid search: linear regression, random forest, XGBoost, neural network
4. Test PCA to reduce the number of features
5. Engineer new features, then average and stack the best models

Hold-out RMSE by model (lower is better)



What the results say

- PCA nudged cross-validated scores up slightly, but hold-out RMSE got worse for every model
- Engineered features (total square footage, property age, garage and second-floor flags) helped the neural network most: 0.167 down to 0.135
- The final stack of linear regression, XGBoost, and a neural network landed near 0.135, similar to the best single models
- On this one split, plain linear regression scored 0.132, so the gains from the extra steps are modest

What I would do next

- Compare all setups with repeated cross-validation, not a single split
- Keep PCA only if speed matters, since it did not improve hold-out accuracy
- Check the neural network and the stack for overfitting before trusting the lowest scores
- Add more engineered features and test them on the simpler models too